

26

QUESTION PAPER
SERIES CODE

A

Registration No. :

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Centre of Exam. : _____

Name of Candidate : _____

Signature of Invigilator _____

ENTRANCE EXAMINATION, 2016

M.Phil./Ph.D. & M.Tech./Ph.D.
COMPUTER & SYSTEM SCIENCES

[Field of Study Code : COMP—SCSP (158)/MTCP (157)]

Time Allowed : 3 hours

Maximum Marks : 480

Weightage : 100

INSTRUCTIONS FOR CANDIDATES

Candidates must read carefully the following instructions before attempting the Question Paper :

- Write your Name and Registration Number in the space provided for the purpose on the top of this Question Paper and in the Answer Sheet.
- Please darken the appropriate Circle of Question Paper Series Code on the Answer Sheet.**
- All questions are compulsory.
- Answer all the 120 questions in the Answer Sheet provided for the purpose by darkening the correct choice, i.e., (a) or (b) or (c) or (d) with BALLPOINT PEN only against each question in the corresponding Circle. Any overwriting or alteration will be treated as wrong answer.
- Each correct answer carries 4 marks. **There will be negative marking and 1 mark will be deducted for each wrong answer.**
- Answer written by the candidates inside the Question Paper will not be evaluated.
- Pages at the end have been provided for Rough Work.
- Return the Question Paper and Answer Sheet to the Invigilator at the end of the Entrance Examination. **DO NOT FOLD THE ANSWER SHEET.**

INSTRUCTIONS FOR MARKING ANSWERS

- Use only Blue/Black Ballpoint Pen (do not use pencil) to darken the appropriate Circle.
- Please darken the whole Circle.
- Darken ONLY ONE CIRCLE for each question as shown in the example below :

Wrong	Wrong	Wrong	Wrong	Correct
 (b) (c) 	 (b) (c) (d)	 (b) (c) 	 (b) (c) 	(a) (b) (c) 

- Once marked, no change in the answer is allowed.
- Please do not make any stray marks on the Answer Sheet.
- Please do not do any rough work on the Answer Sheet.
- Mark your answer only in the appropriate space against the number corresponding to the question.
- Ensure that you have darkened the appropriate Circle of Question Paper Series Code on the Answer Sheet.**

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1. In a class A subnet, the IP address of one of the hosts and the mask are as given below :

IP address : 25.34.12.56

Mask : 255.255.0.0

What is the first address (network address)?

- (a) 25.34.12.0
 - (b) 25.34.12.56
 - (c) 25.34.0.0
 - (d) 25.0.0
2. If every element of a group G is its own inverse, then G is
- (a) finite group
 - (b) abelian group
 - (c) infinite group
 - (d) non-abelian group
3. Which of the following options is the closest in meaning to the sentence below?
"She enjoyed herself immensely at the party"
- (a) She had a terrible time at the party
 - (b) She had a horrible time at the party
 - (c) She had a terrifying time at the party
 - (d) She had a terrific time at the party
4. Consider the two given languages $A = \{x : x \text{ is an integer and divisible by } 2\}$ and $B = \{2x : x \text{ is an integer}\}$. Which of the following is correct?
- (a) A is accepted by an automata, but not B
 - (b) B is accepted by an automata, but not A
 - (c) Both A and B are accepted by an automata
 - (d) None are accepted by an automata

5. What is the maximum value of xy subject to $x + y = 8$?
- (a) 8
 - (b) 16
 - (c) 20
 - (d) 24
6. In which of the following page replacement algorithms, Belady's anomaly may occur?
- (a) FIFO (First In First Out)
 - (b) OPT (OPTimal)
 - (c) LRU (Least Recently Used)
 - (d) MRU (Most Recently Used)
7. What is the primary objective of formal technical reviews?
- (a) To find errors during the process so that they do not become defects after release of the software
 - (b) To find errors after the process so that they do not become defects after release of the software
 - (c) To identify processes that lead to defects after release of the software
 - (d) To identify people who are responsible for defects after release of the software
8. In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?
- (a) He wants all the money
 - (b) He will return the money
 - (c) He will resist all enquiries
 - (d) He will assume final responsibility

9. Of the given choices, what is the output of

$$\sqrt{\frac{-49}{25}} \times \sqrt{\frac{-25}{49}} = ?$$

(a) $\frac{-7}{20}$

(b) $\frac{7}{20}$

(c) 1

(d) -1

10. A linear transformation of T of X into Y is continuous, then which of the following is true?

(a) T is bounded

(b) T is not bounded

(c) T is constant

(d) None of the above

11. Which among the following is a true statement?

(a) For a given finite set of objects, self-hidden surface removal algorithm will remove all the hidden faces

(b) Bezier curves interpolate the given set of control points

(c) To have a steady picture on the screen in case of refreshing display devices the picture has to be refreshed at least 30 times per second

(d) A sphere can be generated by rotating a circle about a specified axis

12. If $x = \sin \theta \sqrt{\cos 2\theta}$, $y = \cos \theta \sqrt{\sin 2\theta}$, then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ will be

(a) 1

(b) -1

(c) 0

(d) Not defined

13. Which of the following is correct for the letter series if the number of letters skipped in between adjacent letters in the series is odd?
- BDHLR
 - EIMQV
 - FIMRX
 - MPROX
14. If $x = \frac{2\sin\theta}{1 + \cos\theta + \sin\theta}$ then what is the value of $\frac{1 - \cos\theta + \sin\theta}{1 + \sin\theta}$?
- $1 + x$
 - $1 - x$
 - x
 - $1/x$
15. How many $32\text{ k} \times 1$ RAM chips are needed to provide a memory capacity of 256 kB?
- 8
 - 32
 - 64
 - 128
16. If two dice are rolled, then what is the probability that their sum is a prime number?
- $\frac{5}{12}$
 - $\frac{5}{6}$
 - $\frac{5}{36}$
 - $\frac{2}{5}$
17. In order to find the names and addresses of Auditoriums with maximum capacity, which of the options given below is needed to complete the given partial SQL query?
- SELECT A1.name, A1.address FROM Auditorium A1*
- WHERE A1.capacity >= All (select A2.capacity from Auditorium A2)
 - WHERE A1.capacity >= Any (select A2.capacity from Auditorium A2)
 - WHERE A1.capacity > All (select max (A2.capacity) from Auditorium A2)
 - WHERE A1.capacity > Any (select max (A2.capacity) from Auditorium A2)

18. Which of the following sentences is grammatically incorrect?

- (a) He is of Asian origin
- (b) She is an European
- (c) They belonged to Africa
- (d) They migrated from India to Australia

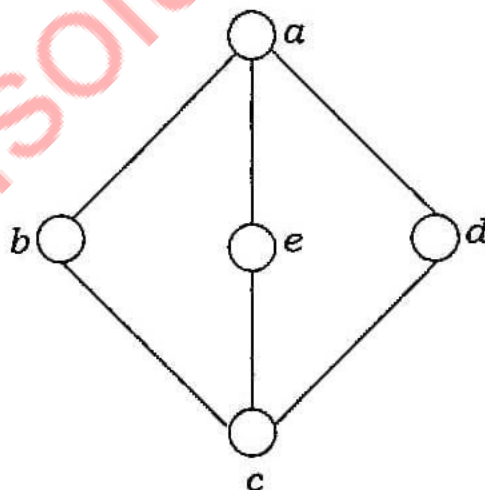
19. The solution of

$$\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$$

satisfying $y(1) = 1$ is given by

- (a) a system of parabolas
- (b) a system of circles
- (c) $y^2 = x(1+x) - 1$
- (d) $(x-2)^2 + (y-3)^2 = 5$

20. Identify the characteristics of the poset $[\{a, b, c, d, e\} \leq 1]$ represented by the following Hasse diagram :



- (a) Not a lattice
- (b) Lattice but not a distributive lattice
- (c) Distributive lattice but not a Boolean algebra
- (d) A Boolean algebra

21. The velocity components along the x and y axes of a moving particle are $-ky$ and kx where k is a non-zero constant. The equation of its path, if it passes through the point $(3, 4)$ is

- (a) $y^2 = x + 5$
- (b) $y = x^2 + 5$
- (c) $xy = 5$
- (d) $y = 5x$

22. Let $R = \{A, B, C, D, E\}$ be a relation scheme with the following dependencies :

$$C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$$

Which of the following is a key for R ?

- (a) CD
- (b) EC
- (c) AE
- (d) AC

23. Which of the following pair of words exhibits the same relationship between each other as the given pair of words?

DERMATOLOGIST : SKIN

- (a) Astrologist : Future
- (b) Paleontologist : Status
- (c) Genealogist : Genes
- (d) Cardiologist : Heart

24. What is the value of $\frac{d}{dx} [\tan^{-1}(\cot x) + \cot^{-1}(\tan x)]$?

- (a) 3
- (b) -2
- (c) 1
- (d) 0

25. For what type of languages, heap allocation is required?

- (a) That support recursion
- (b) That support dynamic data structure
- (c) That use dynamic scope rules
- (d) None of the above

26. Which of the following is true for backward difference operator?

- (a) $\nabla^2 f(x) = f(x - 2h) - 2f(x - h) + f(x)$
- (b) $\nabla^2 f(x) = f(x - 2h) + 2f(x - h) + f(x)$
- (c) $\nabla^2 f(x) = f(x - 2h) - 2f(x - h) - f(x)$
- (d) None of the above

27. If standard deviations for two variables X and Y are 3 and 4 respectively and their covariance is 8, then what is the correlation coefficient between them?

- (a) $\frac{2}{3}$
- (b) $\frac{8}{3\sqrt{2}}$
- (c) $\frac{9}{8\sqrt{2}}$
- (d) $\frac{2}{9}$

28. What for the Banker's algorithm is used in OS?

- (a) Deadlock prevention
- (b) Deadlock avoidance
- (c) Deadlock detection
- (d) Deadlock cure

29. Let $g(x)$ be the inverse of the function $f(x)$ and $f'(x) = \frac{1}{1+x^3}$, then which of the following is equal to $g'(x)$?
- (a) $\frac{1}{1+(g(x))^3}$
- (b) $\frac{1}{1+(f(x))^3}$
- (c) $1+(g(x))^3$
- (d) $1+(f(x))^3$
30. The sum of the marks of four students A, B, C, D is 45. When 2 is added to the marks of A , 2 is subtracted from the marks of B , 2 is multiplied to the marks of C and the marks of D is divided by 2, the teacher get the same number. Who scored the highest marks?
- (a) A
- (b) B
- (c) D
- (d) Data insufficient
31. Consider an undirected graph G with 100 nodes. What is the maximum number of edges to be included in G so that graph is connected?
- (a) 2451
- (b) 4851
- (c) 4950
- (d) 9990
32. If $y = 1 + x + \frac{x^2}{\sqrt{2}} + \frac{x^3}{\sqrt{3}} + \dots + \infty$ then $\frac{dy}{dx}$ is equal to
- (a) x
- (b) y
- (c) 1
- (d) None of the above

33. Let E_1 , E_2 and E_3 be three entities in an E/R diagram with simple single-valued attributes. R_1 and R_2 are two relationships between E_1 and E_2 , where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. R_3 is a one-to-many relationship between E_1 and E_3 which has an attribute of its own. What is the minimum number of tables required to represent this situation in the relational model?
- 3
 - 4
 - 5
 - 6
34. In Picard's method : Given initial value problem $y' \frac{dy}{dx} = f(x, y)$, with $y(x_0) = y_0$, what is the n th approximation of the given problem?
- $y_n = y_0 + \int_{x_0}^x f(x, y_n) dx$
 - $y_n = y_0 + \int_{x_0}^x f(x, y) dx$
 - $y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx$
 - None of the above
35. Choose the most appropriate word from the options given below to complete the following sentence :
- "If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or"
- hyperbolic
 - argumentative
 - restrained
 - indifferent
36. Which of the following is not a standard RS-232C signal?
- RTS
 - CTS
 - DSR
 - VDR

37. What is a compiler-compiler?

- (a) Compiler which compiles a compiler program
- (b) Compiler written in the same language it compiles
- (c) Another name for cross compiler
- (d) Software tool used for automatic generation of a compiler

38. What will be the cyclomatic complexity for the program segment given below?

```
i = 0; n = 4;
while (i < n - 1)
{ j = i + 1;
  while (j < n)
  { if (A[i] < A[j]) swap(A[i], A[j]); }
  i = i + 1;
}
```

- (a) 3
- (b) 4
- (c) 5
- (d) 6

39. Given the function $F = F(x, y, y')$, what the differential equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

is referred as?

- (a) Euler's characteristic equation
- (b) Lagrange's equation
- (c) Hamilton equation
- (d) None of the above

40. For the following statements, choose the correct option based on the conclusions given :

Statements :

- (i) Some books are intelligent
- (ii) No intelligent is wise
- (iii) Some wise are wind

Conclusions :

- I. Some books are not wind
- II. Some books are not wise
- III. Some wind is not intelligent
- IV. Some wise are not books
- (a) Only I and either II or IV follow
- (b) Only III follows
- (c) Only II and III follow
- (d) Only II, III and IV follow

41. What is the value of the determinant

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} ?$$

- (a) 1
- (b) ω
- (c) ω^2
- (d) 0

42. How many digits of the DNIC (Data Network Identification Code) identify the country?

- (a) First three
- (b) First four
- (c) First five
- (d) First six

43. If gcd defines greatest common divisor function, then $\gcd(n_1n_2, n_3n_4) = \gcd(n_1n_3, n_2n_4) = 1$ is possible if and only if

- (a) n_1, n_2, n_3 and n_4 are not pairwise relatively prime
- (b) n_1, n_2, n_3 and n_4 are pairwise relatively prime
- (c) n_1n_2 and n_3n_4 are pairwise relatively prime
- (d) n_1n_2 and n_3n_4 are not pairwise relatively prime

44. Given the following two grammars :

$G1 : S \rightarrow AB \mid bbB$

$A \rightarrow b \mid Ab$

$B \rightarrow a$

$G2 : S \rightarrow bSaS \mid aSbS \mid \lambda$

Which statement is correct?

- (a) Both $G1$ and $G2$ are ambiguous
- (b) $G1$ is ambiguous and $G2$ is unambiguous
- (c) $G1$ is unambiguous and $G2$ is ambiguous
- (d) Both $G1$ and $G2$ are unambiguous

45. How can the number of strongly connected components of a graph may change if a new edge is added in the graph?

- (a) It may increase
- (b) It may decrease
- (c) No change by adding new edge
- (d) None of the above

46. In Newton-Cotes formula, if $f(x)$ is interpolated at equally spaced nodes by a polynomial of degree three then what does it represent?
- (a) Trapezoidal rule
 - (b) Simpson rule
 - (c) Three-eight rule
 - (d) Boole's rule
47. A variable chord is drawn through the origin to the circle $x^2 + y^2 - 2ax = 0$. What is the locus of the centre of the circle drawn on this chord as diameter?
- (a) $x^2 + y^2 + ax = 0$
 - (b) $x^2 + y^2 - ax = 0$
 - (c) $x^2 + y^2 + ay = 0$
 - (d) $x^2 + y^2 - ay = 0$
48. Given following statements about an AVL tree, which set of statements is true?
- I. In this tree heights of two subtrees of every node never differ by more than 1
 - II. Balance factor of each node is -1, 0, 1
 - III. Maximum height of a balanced binary search tree is $1.44 \log_2 4$
- (a) I and II
 - (b) II and III
 - (c) I and III
 - (d) All of the above

49. What is the equation of a plane containing the line of intersection of the planes $x + y + z - 6 = 0$ and $2x + 3y + 4z + 5 = 0$ and passing through $(1, 1, 1)$?
- $20x - 23y + 26z + 69 = 0$
 - $20x - 23y - 26z - 69 = 0$
 - $20x + 23y + 26z + 69 = 0$
 - $20x + 23y + 26z - 69 = 0$
50. Which of the following types of agents can best represent happy and unhappy states?
- Utility based agents
 - Simple reflex agents
 - Learning agents
 - Model based agents
51. There are six students A, B, C, D, E and F participating in an evaluation test for Maths and Science.
- A 's total marks in Maths were just above C and in Science just above F
 - B was just above C in Science but scored less than D in Maths
 - F got more marks than D and E in Science, but didn't perform as well as D in Maths
 - No one scored in between C and D in Maths and C and A in Science
- Which of the following students has scored the least in Science?
- Either D or E
 - Only D
 - Only E
 - A
52. If x' is a bounded linear function on a Hilbert space X , which of the following is true?
- There is a unique $y_0 \in X$ such that $x'(x) = (y_0, x)$, $x \in X$
 - There is a unique $y_0 \in X$ such that $x'(x) = (x, y_0)$, $x \in X$
 - There is a unique $y_0 \in X$ such that $x'(x) = (y_0, x)$, $x \in X$
 - None of the above

53. Which among the following specifies a critical section?

- (a) A region prone to Deadlock
- (b) A piece of code which only a finite number of processes executes
- (c) A piece of code which only one process executes at a time
- (d) None of the above

54. If $a, b, \dots \in G$ a group, and $(abc \dots z)^{-1} = z^{-1} \dots c^{-1} b^{-1} a^{-1}$, then what is this law known as?

- (a) Reversal rule
- (b) Closure rule
- (c) Associative rule
- (d) None of the above

55. There are seven cities A, B, C, D, E, F and G . The following cities have a two-way connection (people can move in both the directions) between them : D and F , E and H , A and B , B and C , G and E , C and E and G and F . There is a one-way connection (people can move only in one way) from F to B .

If a person wishes to travel from D to E , then what is the minimum number of cities that he needs to cross enroute?

- (a) 2
- (b) 3
- (c) 4
- (d) 5

56. What is the value of

$$\int_0^{16} \frac{x^{1/4} dx}{1 + \sqrt{x}} ?$$

- (a) $\frac{4}{3} + \tan^{-1} 2$
- (b) $\frac{8}{3} + 4 \tan^{-1} 2$
- (c) $\frac{2}{3} + \tan^{-1} 2$
- (d) $\frac{5}{3} + \tan^{-1} 2$

57. In which of the following cases, is it possible to obtain different results for call-by-reference and call-by-name parameter passing techniques?

- (a) Passing an expression as a parameter
- (b) Passing an array as a parameter
- (c) Passing a pointer as a parameter
- (d) Passing an array element as a parameter

58. What is the degree of the differential equation

$$\sqrt{\left[\left(\frac{dy}{dx}\right)^2 + \frac{d^2y}{dx^2}\right]} = x ?$$

- (a) 1
- (b) 2
- (c) $\frac{1}{2}$
- (d) 0

59. There are six movies A, B, C, D, E and F to be showed in a film festival. B, C and E are art movies and others are commercial movies. Only F and D are Hindi movies and remaining are English movies. Movies A, C and D are made by Jagmohan Mundhra and movies B, E and F are made by Steven Spielberg.

Which of the English movies is/are art movies made by Steven Spielberg?

- (a) B
- (b) B and C
- (c) C
- (d) B and E

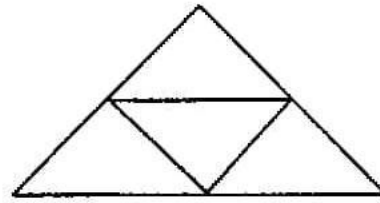
60. Which of the following is not true?

- (a) Class of all languages is not countable
- (b) Every language in P is also in NP
- (c) Every language in NP is decidable
- (d) There are some languages in NP but not in P

61. If $f(x) = \frac{x}{\sqrt{x+1} - \sqrt{x}}$ be a real valued function, then which of the following is true?

- (a) $f(x)$ is continuous, but $f'(x)$ does not exist
- (b) $f(x)$ is differentiable at $x = 0$
- (c) $f(x)$ is not continuous at $x = 0$
- (d) $f(x)$ is not differentiable at $x = 0$

62. For the graph given below, how many spanning trees are possible?



- (a) 24
- (b) 34
- (c) 44
- (d) 54

63. Six people P, Q, R, S, T and U are X 's husband, sister, son, daughter, mother and aunt, though not necessarily in that order. They satisfy the following conditions :

- (i) R and S are either both males or both females
- (ii) Either P or Q , or both are male
- (iii) Either T or U , or both are female

If Q is U 's daughter, then which of the following statements could be true?

- (a) P is X 's sister
- (b) U is R 's aunt
- (c) T is S 's nephew
- (d) Q is S 's niece

64. If x_1 and x_2 are two distinct roots of the equation $a \cdot \cos x + b \cdot \sin x = c$ then $\tan \frac{x_1 + x_2}{2}$ is equal to
- (a) $\frac{a}{b}$
 - (b) $\frac{b}{a}$
 - (c) $\frac{c}{a}$
 - (d) $\frac{a}{c}$
65. Which one of the following is not a storage class supported by C++?
- (a) Register
 - (b) Auto
 - (c) Dynamic
 - (d) Mutable
66. If $x^2 + y^2 = 1$ then what will be the value of $(1 + x + iy) + (1 + x - iy)$?
- (a) $x - iy$
 - (b) $-x - iy$
 - (c) $-x + iy$
 - (d) $x + iy$
67. Two out of Anil, Sunil and Madan are fighting each other. The given statements are
- (i) The shorter one out of Anil and Sunil is the older of the two fighters
 - (ii) The younger one out of Sunil and Madan is the shorter of the two fighters
 - (iii) The taller one out of Anil and Madan is the younger of the two fighters
- Who is the tallest?
- (a) Anil
 - (b) Sunil
 - (c) Madan
 - (d) Data inconsistent